

Quiz (Habanero)

- Q1.** What does the box in a box-and-whisker plot represent?
(A) The range of the data
(B) The interquartile range
(C) The median and quartiles
(D) The mean and mode
(E) The standard deviation
- Q2.** If $Q_1 = 10$ and $Q_3 = 15$, what is the interquartile range?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q3.** A dataset has values 2, 4, 6, 8, 10. What is the median?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q4.** What is an outlier in a dataset?
(A) A value that is close to the median
(B) A value that is within the interquartile range
(C) A value that is more than $1.5 \cdot IQR$ away from Q_1 or Q_3
(D) A value that is equal to the mean
(E) A value that is equal to the mode
- Q5.** The temperature in a city was recorded for 7 days: 22, 25, 28, 30, 32, 35, 40. What is Q_3 ?
(A) 30
(B) 32
(C) 35
(D) 36
(E) 38
- Q6.** What does the whisker in a box-and-whisker plot represent?
(A) The range of the data
(B) The interquartile range
(C) The median and quartiles
(D) The minimum and maximum values
(E) The standard deviation
- Q7.** A box-and-whisker plot shows $Q_1 = 12$, $Q_3 = 18$, and a minimum value of 8. Is 8 an outlier?
(A) Yes
(B) No
(C) Maybe
(D) Need more information
(E) Cannot determine
- Q8.** What is the purpose of a box-and-whisker plot?
(A) To show the correlation between two variables
(B) To compare the distribution of two datasets
(C) To display the mean, median, and mode of a dataset
(D) To visualize the dispersion of a dataset
(E) To calculate the standard deviation

Open Ended Questions (FRQ)

- Q9.** The amount of rainfall in a region was recorded for 10 days: 2, 4, 6, 8, 10, 12, 15, 18, 20, 25. Construct a box-and-whisker plot and identify any outliers. Show your work and explain your answer.
- Q10.** Compare the box-and-whisker plots of two datasets: Dataset A = 1, 3, 5, 7, 9 and Dataset B = 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. How do the plots differ? What do the differences indicate about the datasets? Show your work and explain your answer.

Chef's Tips

- Tip 1: Emphasize the importance of accurate graphing
- Tip 2: Encourage students to check for outliers
- Tip 3: Remind students to consider the context of the problem